Multiple choice Question with one option correct **Q.01 to Q.15**

Multiple choice Question with one or more than one may be correct **Q.16 to Q.35**

Numerical Based Question (no negative marking) **Q.36 to Q.55**

DPP Syllabus : Mole & Equivalent concept, Gaseous state, Chemical equilibrium & Ionic equilibrium

1. For the given reaction :

$2Ag^{+}(aq.) + Cu(s) \rightarrow Cu^{2+}(aq.) + 2Ag(s)$

The equilibrium concentration of Cu^{2+}(aq) is 'x' M. To 500 ml of the above equilibrium solution if 500 ml of water is added then the new equilibrium concentration of Cu^{2+}(aq.) will be -

(A) x M
(B) Less than $\frac{'x'}{2}$ M
(C) More than $\frac{'x'}{2}$ M
(D) $\frac{'x'}{2}$ M

2. For the reaction at equilibrium :

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

On addition of an inert gas at constant pressure & constant temperature, which of the following changes is **NOT** observed ?

(A) the equilibrium constant does not change
(B) concentration of NH_3 decreases
(C) the rate of backward reaction becomes more than the rate of forward reaction just after the addition of the inert gas.
(D) concentration of N_2 increases

3. 2 litres of an acidified solution of $KMnO_4$, containing 1.58 g of $KMnO_4$ per litre, is decolourised by passing sufficient amount of SO_2 gas. If whole of the sulphur from x g of FeS_2 is converted into SO_2 to be used in above reaction, calculate the value of x :-

(A) x = 1.5
(B) x = 3
(C) x = 4.5
(D) x = 6

4. A certain volume of hydroxyl amine (NH_2OH) solution was boiled with an excess of $FeCl_3$ solution to cause the reduction of Fe^{3+} ions according to the reaction :

$Fe^{3+} + NH_2OH \rightarrow Fe^{2+} + N_2O + H^{+} + H_2O$

The resulting solution was estimated for Fe^{2+} ions with 0.5 M $K_2Cr_2O_7$ solution in acidic medium. If the volume of $K_2Cr_2O_7$ solution and NH_2OH solution used are found to be equal, what is the molarity of NH_2OH solution?

(A) 1.5 M
(B) 3 M
(C) 0.75 M
(D) 1 M

5. In 1L saturated solution of CuCl, 0.1 mol AgCl is added. K_{sp} of AgCl = 1.6×10^{-10}. If the resultant concentration of Ag^+ in the solution is 1.6×10^{-7} M, then K_{sp} for CuCl is :-

(A) 1.6×10^{-3} (B) 10^{-3} (C) 10^{-6} (D) 1.6×10^{-6}

6. 40 mL of 0.35 M NaOH solution is added to 50 mL of 0.6 N H_3PO_4 solution. The pH of the mixture would be about : (K_{a1}, K_{a2} and K_{a3} are 10^{-3}, 10^{-8} and 10^{-12} respectively).

(A) 11.82 (B) 3.6 (C) 12.18 (D) 7.82

7. 10 ml of gaseous hydrocarbon is exploded with 100 ml oxygen. The residual mixture on cooling is found to measure 95 ml, of which, 20 ml is absorbed by KOH and remaining by alkaline pyrogallol. It is known that alkaline pyrogallol absorbs O_2. Predict the formula of hydrocarbon :

(A) C_2H_6 (B) C_2H_4 (C) C_2H_2 (D) C_4H_8

8. If the number of molecules of SO_2 (Mol. weight = 64 u) effusing through an orifice of unit area of cross section in unit time at 0°C and 1 atm pressure is n, the number of He molecules (atomic weight = 4 u) effusing under similar conditions at 273°C and 0.25 atm is :

(A) $\frac{n}{\sqrt{2}}$ (B) $n\sqrt{2}$ (C) 2n (D) $\frac{n}{2}$

9. Statement-1 : If KIO_3 reacts with excess KI in acidic medium, and the produced I_2 is titrated with hypo solution, then milliequivalents of KIO_3 used and hypo used are equal.

Statement-2 : According to law of equivalence, in a chemical reaction, milliequivalents of all the reactants are equal and also equal to milliequivalents of each product.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

10. Statement-1 : The pH of a 0.003 M aqueous solution of NH_4CN can be approximately calculated using the formula : $pH = \frac{1}{2}\left(pK_W + pK_a - pK_b\right)$ Given : K_a(HCN) = 4×10^{-10} & $K_b(NH_3) = 2 \times 10^{-5}$

Statement-2 : The degree of hydrolysis (h) of NH_4CN in its 0.003 M aqueous solution comes out to be greater than 0.1 and so, its value cannot be neglected with respect to 1 .

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

11. Calculate the volume (in mL) of 0.1 M Na_2SO_4 solution, which must be added to 10 mL of HCl solution (pH = 1), so that pH of the resulting solution becomes 2. (Given : K_{a_2} for $H_2SO_4 = 10^{-2}$ and $K_{a_1} = \infty$)

(A) 5 (B) 10 (C) 15 (D) 20

12. 0.2 millimoles of Zn^{2+} ion is mixed with $(NH_4)_2S$ solution of molarity 0.02 M. The amount of Zn^{2+} that remains unprecipitated in 20 mL of this solution would be : (Given : K_{SP} ZnS = 4×10^{-24})

(A) 5.2×10^{-22} g (B) 2.6×10^{-22} g

(C) 1.04×10^{-21} g (D) 5.2×10^{-23} g

13. At low pressure if RT = $2\sqrt{aP}$ then volume occupied by van der Waals gas is :

(A) $\frac{2RT}{P}$ (B) $\frac{2P}{RT}$ (C) $\frac{RT}{2P}$ (D) $\frac{2RT}{a}$

14. A spherical air bubble is rising from the depth of a lake when pressure is p atm and temperature is T K. The percentage increase in its radius when it comes to the surface of a lake will be (Assume temperature and pressure at the surface to be 2T K and p/4, respectively).

(A) 100% (B) 50% (C) 25% (D) 200%

15. An unknown gaseous hydrocarbon and oxygen gas are mixed in volume ratio 1 : 7 respectively and exploded. The resulting mixture upon cooling occupied a volume of V mL, half of which got absorbed in aq. KOH (absorbs CO_2) and the remaining half was absorbed in alkalline pyrogallol solution (absorbs O_2). Assuming all volumes to be measured under identical conditions :

(A) The molecular formula of hydrocarbon can be C_2H_6.

(B) The resulting mixture obtained after cooling contains 50% O_2 and 50% CO by mole.

(C) The molecular formula of hydrocarbon can be C_3H_4.

(D) Molecular mass of hydrocarbon is definitely less than 28 u.

16. Choose the correct option(s) -

(A) Favourable conditions for formation of graphite are high pressure and low temperature from equilibrium diamond (d = 3.5 gm/ml) $\rightleftharpoons$ graphite (d = 2.3 gm/ml) ; $\Delta H = -1.9$ kJ/mole

(B) For reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$; degree of dissociation (α) is $\alpha = \sqrt{\frac{K_P}{4P + K_p}}$ where P is equilibrium pressure and equilibrium has been established starting from N_2O_4.

(C) For reaction $Cl_2(g) + 3F_2(g) \rightleftharpoons 2ClF_3(g)$; $\Delta H = -329$ kJ, dissociation of $ClF_3(g)$ will be favoured by adding of inert gas at constant pressure and temperature.

(D) Reaction stops at equilibrium (microscopically)

17. On heating a mixture of SO_2Cl_2 & CO, two equilibrium are simultaneously established :

$SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

$CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$

On adding SO_2 at equilibrium, what will happen at new equilibrium wrt previous equilibrium ?

(A) $[SO_2Cl_2]$ decreases (B) [CO] increases

(C) $[Cl_2]$ decreases (D) $[SO_2]$ decreases

18. At 263K, a solid compound $A(H_2O)_8$ shows following equilibria (Assume 263K is sublimation point of $H_2O(s)$ under given condition) -

$A(H_2O)_8(s) \rightleftharpoons A(g) + 8\,H_2O(g)$; Kp_1

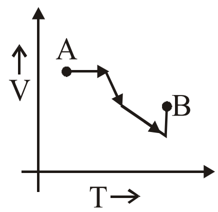
$A(H_2O)_8(s) \rightleftharpoons A(g) + 8\,H_2O(s)$; Kp_2

If equilibrium partial pressures of two gases A & H_2O are 0.2 bar & 0.001 bar respectively, then correct information is/are -

(A) $K_{p_1} = 2\times10^{-25}\,bar^9$ (B) $K_{p_2} = 0.2$ bar

(C) Vapour pressure of ice is 0.001 bar (D) Vapour pressure of ice is 0.002 bar

19. For the following V-T plot for an ideal gas undergoing a process from state A to state B, select the correct alternative(s).



(A) Pressure constantly increases

(B) Pressure first increases, then decreases

(C) Final pressure is more than initial pressure

(D) Pressure first decreases then increases.

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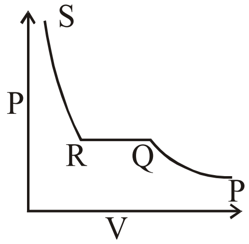
20. The above isotherm was observed for a monoatomic gas at certain temperature. Which of the following is/are correct ?



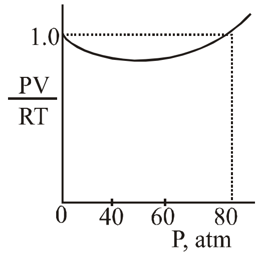
(A) The gas is behaving ideally

(B) The gas is below its Boyle's temperature

(C) In the horizontal QR, the pressure is more than critical pressure

(D) The gas shows negative deviation from ideal gas in PQ and QR.

21. The van der Waals equation of state for a non-ideal gas can be rearranged to give $\frac{PV}{RT} = \frac{V}{V-b} - \frac{a}{VRT}$ for 1 mole of gas. The constants a & b are positive numbers. When applied to H_2 at 80K, the equation gives the curve as shown in the figure. Which of the following statements is(are) correct :



(A) The two terms $\frac{V}{(V-b)}$ and $\frac{a}{VRT}$ are never equal

(B) At 80 atm, the two terms $1 + \frac{V}{(V-b)}$ & a/VRT are equal.

(C) At a pressure greater than 80 atm, the term V/(V – b) is greater than $1 + \frac{a}{VRT}$.

(D) At 60 atm, the term V/(V – b) is smaller than $1 + \frac{a}{VRT}$.

22. 50 ml of 20.8% w/v $BaCl_2$ (aq) and 100 ml of 9.8% w/v H_2SO_4(aq) solution are mixed. Then in final solution :

(A) $[Cl^-] = 0.66$ M (B) $[H^+] = 1.33$ M (C) $[Ba^{2+}] \approx 0$ M (D) $[SO_4^{2-}] = 0.33$ M

23. The volume of HCl solution of specific gravity 1.2 and 3.65% by weight, which can produce atleast 1.12 L Cl_2 at 1 atm and 273 K by the following reaction -

$$MnO_2 + HCl \rightarrow MnCl_2 + H_2O + Cl_2$$

(A) 200 ml (B) 166.7 ml (C) 333.3 ml (D) 267 ml

24. The pH of a 0.01 M acid HX is 2 and pH of 0.01 M salt ACl is also 2. What conclusions **cannot** be drawn from this information?

(A) HX is a weak acid and AOH is strong base.

(B) HX is a strong acid and AOH is strong base.

(C) HX is a strong acid and AOH is very weak base.

(D) HX is a strong acid and A^+ undergoes partial hydrolysis.

25. For the reaction 'p' HNO_2 + 'q' $KMnO_4$ + 'r' $H_2SO_4 \longrightarrow$'s' HNO_3 + 't' $MnSO_4$ + 'u' $K_2SO_4 + 3H_2O$

Which of the following is/are true ?

(A) H_2SO_4 is reducing agent

(B) HNO_2 is reducing agent

(C) $p + q + r = 10$

(D) Equivalent weight of HNO_3 in the reaction is $\frac{\text{Molar mass}}{2}$

26. The pH of 0.1 M solution of a weak base is 11. On diluting the solution, select the **INCORRECT** statement(s):

(A) pH increases

(B) $[OH^-]$ increases

(C) α decreases

(D) Number of H^+ ions in solution increases

27. Which is/are correct statement(s) ?

(A) $H_2PO_2^-$ and HCO_3^- are amphiprotic species

(B) Equivalent weight of H_3PO_4 can be equal to its molar mass depending on the reaction.

(C) $KMnO_4$ has maximum equivalent weight in acidic medium as oxidising agent.

(D) Oxidation state of H in H_2 is more than that in NaH

28. Consider the following statements :

(I) An increase in pressure (caused by decrease in volume) at equilibrium results in increase in molar concentration of each gaseous substance involved.

(II) An increase in pressure (caused by decrease in volume) at equilibrium results in increase in no. of moles of each gaseous substance involved.

(III) For the reaction $A(s) \rightleftharpoons B(g) + C(g)$ starting with A only at constant temperature, total pressure at equilibrium (of B(g) & C(g)) is not affected by changing the volume of container.

(IV) $H_2O(l) \rightleftharpoons H_2O(g)$; on temperature increase, Kp of this reaction is increased.

Tick the correct statement(s)

(A) (I)

(B) (II)

(C) (III)

(D) (IV)

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29. For a reaction : $aA + bB \longrightarrow cC + dD$

Three students stated different ways of determining limiting reagent.

Student 1 : Calculate the minimum moles of 'A' needed to completely consume 'B', and if available amount of 'A' exceeds what is needed, then 'B' is limiting reagent otherwise 'A' will be limiting reagent.

Student 2 : Calculate the ratio of the moles of the reactants initially taken, then compare it to theoretical mole ratio(according to stochiometry of the reaction). If the theoretical ratio exceeds ratio of moles actually taken, then reactant in numerator will be limiting reagent.

Student 3 : Calculate the amount of product (any one of the product) that can be obtained if each reactant is completely consumed and that reactant is limiting reagent which has produced least mass of product.

Assume that atleast one of A or B is the limiting reagent.

(A) Student-1 has defined limiting reagent correctly.

(B) Student-2 has defined limiting reagent correctly.

(C) Student-3 has defined limiting reagent correctly.

(D) If Student 3 in first experiment finds that when 1 mole of 'A' reacted with excess of reagent 'B' and in second experiment when 1 mole of 'B' reacted with excess of reagent 'A', then in the later experiment mass of the product produced was greater. Then A should be the limiting reagent when 1 mole each of A and B are reacted.

30. What will **not** happen when $FeCl_3$ is added to a suspension of $Al(OH)_3$?

K_{sp} of $Cr(OH)_3$, $Al(OH)_3$ and $Fe(OH)_3$ are 7×10^{-31}, 2×10^{-33} and 6.4×10^{-38} respectively.

(A) Colour of suspension remains same

(B) Colour of suspension changes to reddish brown

(C) Precipitate of $AlCl_3$ will be produced

(D) $FeCl_3$ will not show any reaction

31. Select the correct match(es) :

	Column-I		Column-II
(1)	50 mL of 3M HCl solution + 50 mL of 1M $ZnCl_2$ solution	p	4.17 m
(2)	An aqueous solution of NaCl with mole fraction of NaCl as 0.1	q	$[Cl^-] = 3$ M
(3)	20% (w/w) propanol (C_3H_7OH) aq. solution	r	$[H^+] = 0.75$ M
(4)	10.95% (w/v) HCl solution	s	6.1 m

(A) (A) – r (B) (B) – s (C) (C) – p (D) (D) – q

32. Which of the following solutions will have $6 < pH < 10$?

Given that :

$K_a\ (CH_3COOH) = 2 \times 10^{-5}$, $K_a\ (C_2H_5COOH) = 8 \times 10^{-6}$, $K_a\ (ROH) = 3 \times 10^{-13}$

$K_1\ (H_2CO_3) = 4 \times 10^{-7}$, $K_2\ (H_2CO_3) = 4 \times 10^{-11}$, $\log 2 = 0.3$; $\sqrt{1.12} = 1.06$

(A) 0.1 M CH_3COONa and 0.1 M $(C_2H_5COO)_2Ba$

(B) 0.1 M $NaHCO_3$

(C) 0.1 M (aq.) ROH

(D) 10^{-3} M (aq.) RONa

33. For which of the following, solubility in acidic solution is more than that in pure water ?

Consider no common ion effect from anion of acid.

(A) AgBr (B) AgCN (C) $Fe(OH)_3$ (D) $Zn(OH)_2$

34. When you see the tip of a match flare, the chemical reaction is likely to be -

$P_4S_3 + O_2 \rightarrow P_4O_{10} + SO_2$

Which of the following is/are correct ?

(A) The minimum mass of P_4S_3 required to produce at least 1gm of each of the product is $\frac{220}{284}$ gm

(B) It is an example of comproportionation reaction

(C) The number of electrons lost by one molecule of P_4S_3 is 32.

(D) The number of electrons gain by one molecule of O_2 is 4.

35. Select the correct statement(s) for an ideal gas :

(A) If all gas molecules are assumed to be rigid sphere of negligible volume, the only possible molecular motion is translational.

(B) When temperature increases at constant pressure, collision frequency increases

(C) Mean free path increases by increasing temperature at constant pressure

(D) On increasing the temperature, the fraction of molecules at low speed, increases and the fraction of molecules at high speed, decreases

36. How many maximum moles of $Fe_2(SO_4)_3$ can be added in 2 L water at 25°C without precipitating $Fe(OH)_3$?

K_{sp} of $Fe(OH)_3$ is 6.4×10^{-38} . Report your answer after multiplying by 10^{17}.

37. Zelina, a student of class XI is working in the chemistry lab of her school. She is provided with 4 containers of large capacity by the lab assistant.

Container 1 contains 2L of '2.8 V' H_2O_2.

Container 2 contains 2L of '16.8 V' H_2O_2.

Container 3 contains sufficient amount of water.

Container 4 is empty.

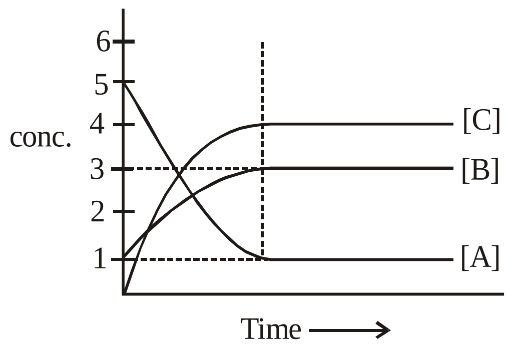
She has been asked by her teacher to prepare H_2O_2 solution using the components of container 1, 2 or 3 (partially or completely) and store it in container 4.

Zelina prepared a 12.6 V H_2O_2 solution and mixed it with excess of KI solution and titrated the liberated I_2 with $\frac{9}{7}$ M hypo solution. Find the maximum volume (in L) of hypo solution that could have been consumed in above process : (Here, volume strength is defined at 1 atm, 273K)

38. Equilibrium constant for the given reaction will be

$aA \rightleftharpoons bB + cC$

(Note : a , b & c are minimum integral stoichiometric coefficients)



39. PCl_5 dissociates into PCl_3 & Cl_2. At equilibrium pressure of 3 atm, all three gases were found to have equal number of moles in a vessel. In another vessel, equimolar mixture of PCl_5, PCl_3 & Cl_2 are taken at the same temperature but at an initial pressure of 9 atm. Then find the partial pressure of Cl_2 (in atm) at equilibrium in second vessel.

40. Find sum of total moles of $KMnO_4$ needed for oxidation of following [i to iv]

(i) 1 mol of FeC_2O_4 in acidic medium

(ii) 1 mol of I^- in basic medium

(iii) 2 moles of FAS in acidic medium

(iv) 2.5 moles Na_3AsO_3 in acidic medium

41. (a) 'a' moles of $K_2Cr_2O_7$ are needed in acidic medium for the oxidation of 9 moles of ethanol to acetic acid.

(b) An unknown metal chloride undergoes reduction reaction with Mg producing metal and $MgCl_2$. Experiments show that 52.4 g of metal chloride reacts with 9.6 g of Mg according to given reaction. The equivalent weight of metal in the given metal chloride is 'b'.

Report your answer as $\left(\frac{b}{a}\right)$ (Metal chloride + Mg $\longrightarrow$ Metal + $MgCl_2$)

42. An organic compound contains C, H and O atoms. One molecule of the compound contains H-atoms equal to 66.67 % of total atoms and mass ratio of C to O is 3 : 2. If the molecular formula of the compound is $C_xH_yO_z$, what is the value of X + Y + Z. (Given : vapour density of compound is 23)

43. 200 mL of 0.1 M aqueous solution of acetic acid is mixed with equal volume of equimolar HCl solution at 27°C. If 1 g of NaOH is added to this, then the $[H^+]$ in final solution is $x \times 10^{-y}$ M (represented in scientific notation). Find x + y. K_a of acetic acid = 2×10^{-5}

44. A certain amount of Dichloroacetic acid ($CHCl_2COOH$) is oxidised to CO_2, H_2O and Cl_2 by 300 gram equivalents of $KMnO_4$ in acidic medium. How many moles of Barium hydroxide are required to completely neutralize the same amount of acid ?

45. If the density of methanol is 0.793 kg L^{-1}, what is its volume (in mL) needed for making 0.0027 m^3 of its 0.22 M solution ?

46. Equilibrium constant for the given reaction is $K = 10^{20}$ at temperature 300 K

$A(s) + 2B(aq.) \rightleftharpoons 2C(s) + D(aq.)$

Calculate the equilibrium concentration of B (in mol/L) starting with mixture of 1 mole of A and 1/2 mole/litre of B in a container of volume 1L at 300 K

(Give your answer by multiplying it with 10^{12})

47. The compression factor (compressibility factor) for 1 mole of a van der Waals gas at 727ºC and 100 atmosphere pressure is found to be 0.5. Assuming that the volume of gas molecule is negligible, calculate the van der Waals constant a (in atm L^2mol^{-2}). **[R = 0.082 atm × litre/mol K]**

Round off your answer to nearest whole number.

48. Acid sample is prepared using HCl, H_2SO_4, H_2SO_3 & H_3PO_4 separately or as mixture of any two or more. Calculate the minimum volume of 4% w/v NaOH added (in ml) to 294 gm sample, in order to ensure complete neutralisation in every possible case.

49. Hardness of water sample is due to only dissolved $MgSO_4$. To start precipitation of MgF_2, the concentration of NaF required is just more than 0.02M. What will be the degree of hardness of water sample (in ppm of $CaCO_3$).

[Given : K_{sp} (MgF_2) at 25°C = 8×10^{-8}]

50. Determine the number of correct statements :

(a) With increase in temperature, neutral pH decreases

(b) At neutralisation point in acid-base titration, solution may be acidic or basic

(c) Solution of CH_3COOH & CH_3COONa show maximum buffer action when molar ratio is 1 : 1, when total moles of acid & salt are constant.

(d) Degree of hydrolysis of every salt decreases on dilution

(e) Phenolphthalin is the suitable indicator for titration of CH_3COOH & NaOH.

51. 100 ml of a solution which is 0.050 M in the acid HA (pK_a = 4) and 0.10 M in HB(pK_a = 8) is titrated with 0.2 M NaOH. Calculate the pH at the first equivalence point

52. A 1 litre solution containing equal moles of FeO and $Fe_{0.8}O$ was titrated with 70 ml, 0.3M $KMnO_4$ in acidic medium. Millimoles of Fe^{3+} produced are -

53. For one mol of a real gas kept in a rigid vessel, the graph of $\frac{P}{T}$ vs P intersects $\frac{P}{T}$ axis at 2. Hence, the volume of the vessel is R/x (R = universal gas constant). Determine x. Report your answer as 0, if you find data insufficient :

54. If specific volume of H_2 gas at 410 atm is 0.2 lit/g, then temperature of the gas (in K) assuming it as van der Waals gas having a = 0.25 dm^6 atm mol^{-2} & b = 0.02 lit/mol is ___approx. (R = 0.082 Latm/K-mol)

55. The ratio of kinetic energy of all the molecule of an ideal gas present in unit volume of closed container and that of pressure exerted by ideal gas sample is –

ANSWER-KEY DPP # 04 (MAIN + ADVANCED)

1.	**(B)**	**2.**	**(D)**	**3.**	**(B)**	**4.**	**(A)**	**5.**	**(C)**
6.	**(D)**	**7.**	**(C)**	**8.**	**(A)**	**9.**	**(D)**	**10.**	**(B)**
11.	**(C)**	**12.**	**(A)**	**13.**	**(C)**	**14.**	**(A)**	**15.**	**(C)**
16.	**(BC)**	**17.**	**(BC)**	**18.**	**(ABC)**	**19.**	**(BC)**	**20.**	**(BD)**
21.	**(ACD)**	**22.**	**(ABCD)**	**23.**	**(ABCD)**	**24.**	**(ABD)**	**25.**	**(BCD)**
26.	**(ABC)**	**27.**	**(BD)**	**28.**	**(CD)**	**29.**	**(ABCD)**	**30.**	**(ACD)**
31.	**(ABCD)**	**32.**	**(ABC)**	**33.**	**(BCD)**	**34.**	**(CD)**	**35.**	**(AC)**
36.	**(6.4)**	**37.**	**(5)**	**38.**	**(48)**	**39.**	**(2)**	**40.**	**(4)**
41.	**(5)**	**42.**	**(9)**	**43.**	**(11)**	**44.**	**(25)**	**45.**	**(24)**
46.	**(50)**	**47.**	**(17)**	**48.**	**(9000)**	**49.**	**(20)**	**50.**	**(4)**
51.	**(5.85)**	**52.**	**(135)**	**53.**	**(2)**	**54.**	**(1900)**	**55.**	**(1.5)**

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